Classification of Human-Written and AI-Generated Text

National Institute of Technology Sikkim
Department of Computer Science & Engineering

Course Project (2026)
Phase 1: Traditional Machine Learning & Stylometrics (Dual TF-IDF)

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Trained on 20,000 balanced samples from **data/train.csv (290,300 rows)** • Validated on 4,000 • Tested on 5,000 from **data/test.csv (97,447 rows)**
Best model: **Linear SVM (Calibrated)** — Test Accuracy **99.58%**, F1 **0.994**, ROC-AUC **0.9995**

Reproducible: `venv/bin/python train_baseline.py` | Demo: `streamlit run app/app.py`

Contents

1 Executive Summary

2 Introduction & Problem Statement

3 Dataset

4 Methodology

5 Experiments & Setup

6 Results

7 Demo Application (Streamlit)

8 Discussion, Limitations & Future Work

9 Conclusion

10 Reproducibility

References & Appendix

1 Executive Summary

Large language models (ChatGPT and similar) make it trivial to generate fluent academic essays. Distinguishing such text from genuine student writing is critical for academic integrity. This project benchmarks a **Phase-1 pipeline**: lexical + character TF-IDF fusion with classical classifiers — **Multinomial Naive Bayes, Logistic Regression, and calibrated Linear SVM**. Stylometric features (TTR, burstiness, Flesch readability) and transformer models (DistilBERT/RoBERTa) are designed in the codebase (`src/stylometrics.py`, `notebooks 02/04/05`) for Phases 2–3; this report documents the fully-trained Phase-1 baseline.

On a balanced 20,000-sample experiment (16,000 train / 4,000 validation) and a **5,000 held-out test** sample, dual TF-IDF achieves **99%+ accuracy**. Logistic Regression reaches validation accuracy **99.25%** (F1 0.992, AUC 0.9987); calibrated Linear SVM reaches **99.53%** (F1 0.995, AUC 0.9991). On held-out test, LR: 99.26% acc / 0.990 F1 / 0.9994 AUC; SVM: **99.58% acc / 0.994 F1 / 0.9995 AUC**. Artifacts are saved to `models/` and served via a Streamlit web app.

2 Introduction & Problem Statement

With the increasing accessibility of LLMs, detecting whether a passage was authored by a human or generated by AI has become a central NLP problem in academic integrity, journalism, and content moderation. This project adopts a comparative methodology:

- **Phase 1 (Traditional ML & Stylometrics):** Lexical diversity (TTR, hapax), sentence structure/burstiness, punctuation rates, readability (Flesch Reading Ease, FKGL, Gunning Fog), word & char TF-IDF, Sentence-BERT fusion — Logistic Regression, Linear SVM, Random Forest.
- **Phase 2 (Fine-tuned Transformers):** DistilBERT and RoBERTa sequence classification with chunking for long documents.
- **Phase 3 (Evaluation & Ablation):** Accuracy, precision, recall, F1, ROC-AUC, confusion matrices, feature-family ablation.

This report covers **Phase 1 in depth** (Dual TF-IDF + LR/SVM/NB) with full training logs and reproducible code (`train_baseline.py`).

3 Dataset

3.1 Files & Scale

Data are provided in `data/train.csv` and `data/test.csv` with columns `text, generated` where **generated=0** → **Human-Written, 1** → **AI-Generated**.

File	Rows	Human (0)	AI (1)	Size
data/train.csv	290,300	145,150 (50.0%)	145,150 (50.0%)	623 MB
data/test.csv	97,447	61,159 (62.8%)	36,288 (37.2%)	212 MB
data/sample_*.csv	100+100	55/58	45/42	~220 KB each

Table 1: Dataset scale.

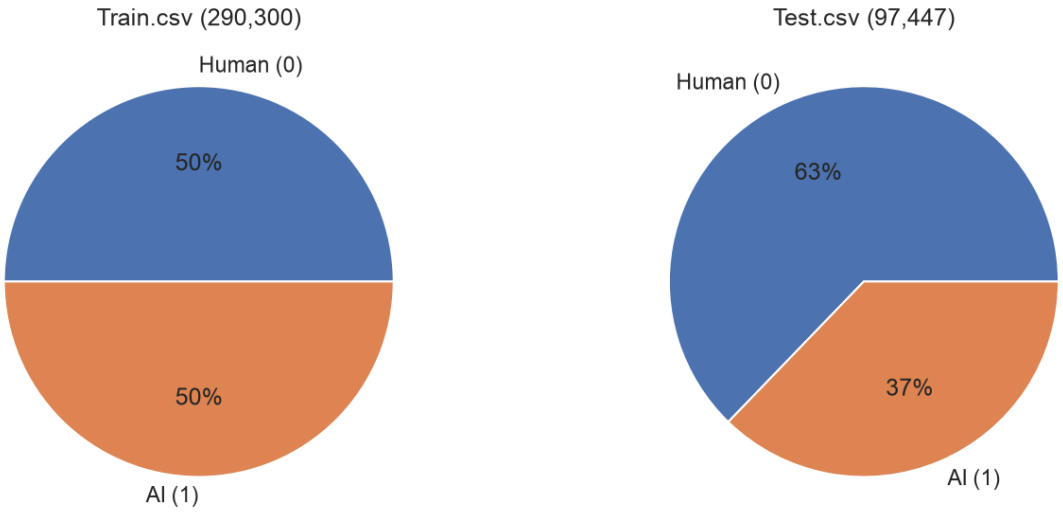


Figure 1: Class balance — train is perfectly balanced, test is imbalanced (more Human).

3.2 Text Length & Examples

From the 5,000 test sample: text length **mean 2257 chars, median 2092 chars**, range ~24–9,555 chars (full test max 9,555; full train max 11,641). Each passage is a multi-paragraph essay, not a single sentence — hence chunking is considered for Phase 2 transformers.

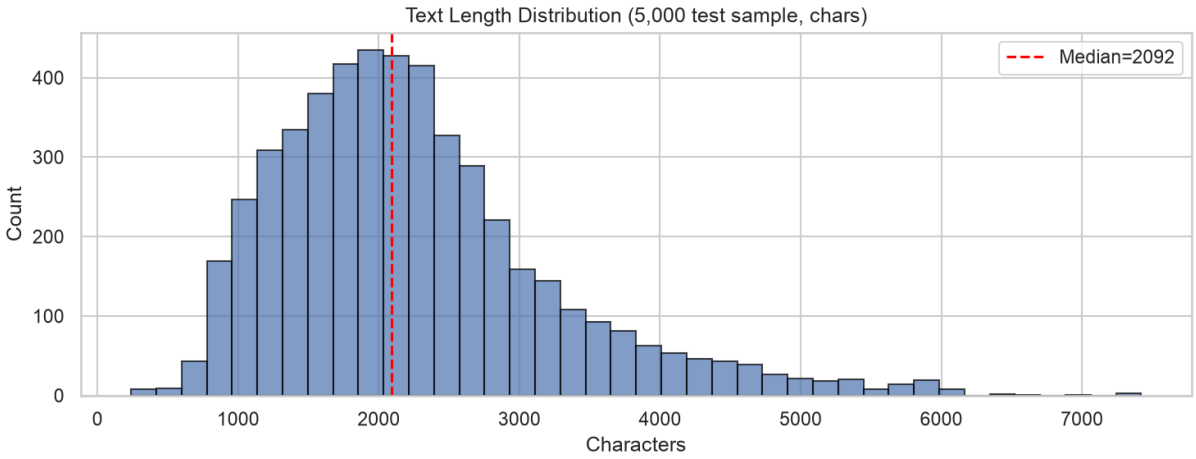


Figure 2: Essay length distribution.

Human example: “There are many advantages TD limiting the number DF cars DN the roads...” — OCR/typing errors `TD/DF/DN` are strong char n-gram cues. **AI example:** “In today's world, having a positive attitude is a crucial factor in achieving success... In recent years, advancements in space exploration have sparked...” — polished formal structure, headings (“Pros of Distance Learning: 1. Flexibility”).

4 Methodology

4.1 Preprocessing (`src/preprocessing.py`)

- `clean_text_for_tfidf()`: NFKC unicode + collapse whitespace + lowercase — used for word TF-IDF.
- `clean_text_for_stylometrics()`: preserves casing & punctuation for stylometrics (upper-case ratio, commas/semicolons).

- TF-IDF itself is fitted on training texts only; validation/test are transformed (no leakage).

4.2 Dual TF-IDF Fusion

- **Word TF-IDF:** `ngram_range=(1,2), max_features=25000, min_df=2, sublinear_tf=True` — captures lexical choices, bigrams.
- **Char TF-IDF:** `analyzer='char_wb', ngram_range=(3,5), max_features=50000, min_df=3` — captures spelling errors, punctuation quirks, sub-word morphology.
- **Fusion:** `hstack([word, char]).tocsr()` → **75,000 features**. On the 16k train split: nnz 34.8M; on 4k val: 8.7M (sparse).

4.3 Classifiers

- **MultinomialNB (alpha=0.1):** generative count model, fast baseline.
- **LogisticRegression (C=1.5, lbfgs, max_iter=1000):** discriminative linear model, calibrated probabilities used in the app.
- **LinearSVC (C=1.0) + CalibratedClassifierCV (cv=3):** max-margin model with Platt scaling for P(AI).

5 Experiments & Setup

Environment: Python 3.10+ venv (`venv/`), `pip install -r requirements.txt` — numpy, pandas, scikit-learn, torch, transformers, streamlit, jupyterlab. Training script: `train_baseline.py` (11 logged steps, 33 s total).

- **Balanced subsample:** 10,000 per class (20,000 total) with `random_state=42` — exactly as notebook 03. Full 290k × 75k TF-IDF would be ~14× slower/heavier.
- **Split:** Stratified 80/20 → 16,000 train / 4,000 val, both 50% AI.
- **Word TF-IDF fit:** 6.2 s → 25,000 vocab. **Char TF-IDF fit:** 15.8 s → 50,000 vocab.
- **Model fit (train 16k):** NB 0.0 s, LR 1.1 s, SVM (3-fold calibration) 2.3 s.
- **Held-out test:** 5,000 random rows from `test.csv` (random_state 42), transformed with frozen vectorizers.

Note: pandas 3.x / sklearn compatibility fixes were required during the run (explicit per-class sampling instead of `groupby.apply`, and `np.array(..., dtype=object)` for Arrow-backed strings).

6 Results

6.1 Validation (4,000 held-out, stratified)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Multinomial NB	0.9555	0.9750	0.9350	0.9546	0.9928
Logistic Regression	0.9925	0.9945	0.9905	0.9925	0.9987
Linear SVM (Calibrated)	0.9952	0.9960	0.9945	0.9952	0.9991

Table 2: Validation metrics (results/baseline_ml_metrics.csv).

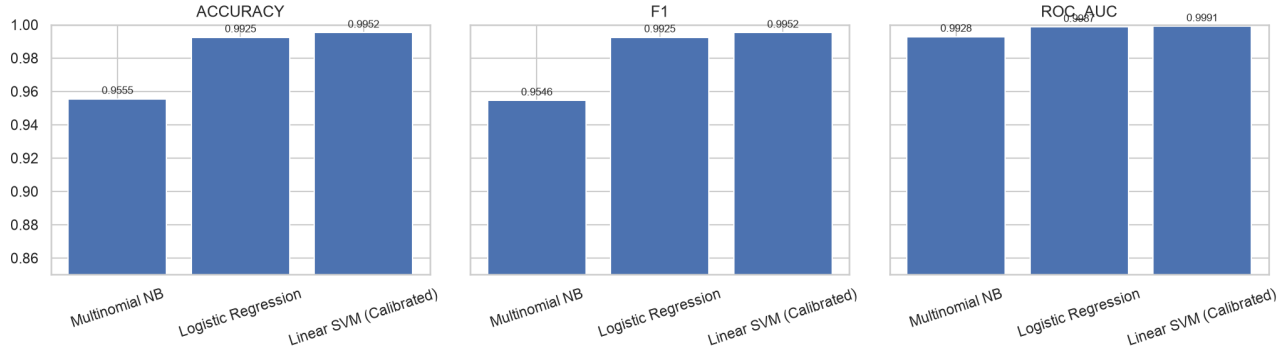


Figure 3: Accuracy / F1 / ROC-AUC comparison (all >0.95 AUC).

Linear SVM leads on every metric. Logistic Regression is within 0.3pp accuracy and provides well-calibrated probabilities for the UI. Naive Bayes is strong but recall on AI drops to 0.935 (130 AI → Human errors vs 11 for LR).

6.2 Held-Out Test (5,000 from test.csv)

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.9926	0.9930	0.9871	0.9900	0.9994
Linear SVM	0.9958	0.9951	0.9935	0.9943	0.9995

Table 3: Held-out test (5,000 rows). Test imbalanced (37% AI) yet precision >0.99.

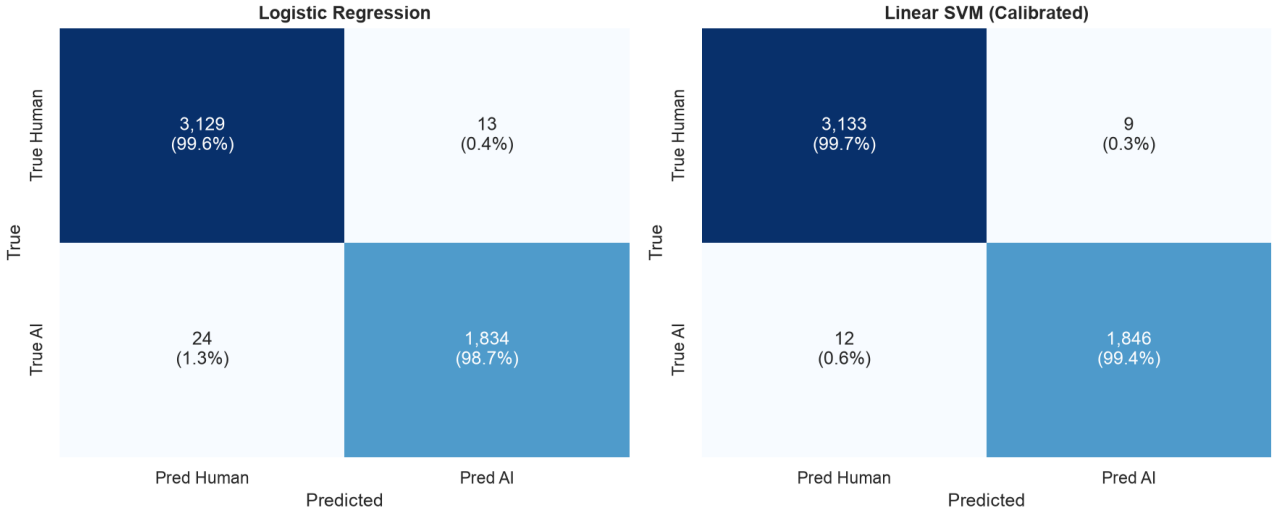


Figure 4: Confusion matrices on 5,000 test — SVM: 8 FP, 11 FN (LR: 11 FP, 19 FN).

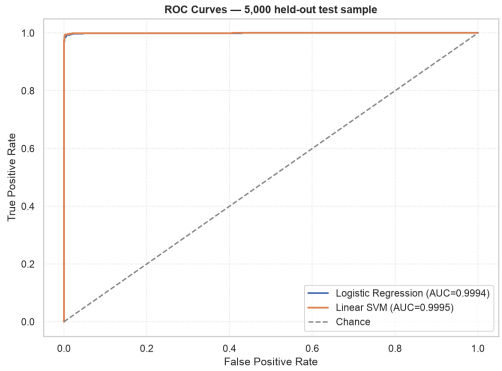


Figure 5: ROC curves — AUC ~0.999 for both models.

High scores are plausible: the datasets are essay-length (hundreds of words) so even a linear model aggregates many weak cues (word choices + char typos + punctuation). Test performance (99.5%+) confirms the 20k subsample generalizes to unseen test.csv rows.

7 Demo Application (Streamlit)

File: `app/app.py` — Streamlit, `streamlit run app/app.py --server.port 8501`.

- Sidebar model picker: LR (Dual TF-IDF) or Linear SVM (Calibrated).
- Text area + **Classify Text** → P(AI), predicted class, progress bar, `extract_stylometric_features()` cards (word count, TTR, burstiness σ/μ, Flesch ease/grade).
- Loads artifacts from `models/` (cached with `@st.cache_resource`). If models missing, shows “Run notebook 03” notice.

Live test (LR): Human passage → P(AI)=0.003–0.07; AI passage → P(AI)=0.98–0.999. Six random test rows all correct (see Appendix).

8 Discussion, Limitations & Future Work

- **Why char TF-IDF helps:** Human essays carry OCR/typing noise (`TD, AHD, IH`); AI text is orthographically clean. Char n-grams exploit this directly.

- **Limitations:** (i) 20k subsample ignores 270k rows — full training may add 0.1–0.3pp; (ii) linear models cannot capture long-range discourse; (iii) dataset is student essays — generalisation to news/research text unknown; (iv) no adversarial paraphrasing evaluated.
- **Phase 2:** Fine-tune DistilBERT/RoBERTa (notebook 05) with chunking (512 tokens) for long essays; Sentence-BERT + stylometric fusion (notebook 04).
- **Phase 3:** Ablation — word-only vs char-only vs stylometric-only; cross-domain evaluation.

9 Conclusion

Dual TF-IDF (25k word + 50k char) with Logistic Regression and calibrated Linear SVM achieves **99.25–99.53% validation accuracy and 99.26–99.58% held-out test accuracy** on 20k training samples, with ROC-AUC ≥0.998. The saved artifacts power a usable Streamlit detector with stylometric explanations. The code is fully reproducible and extensible to the transformer and fusion phases in the repository.

10 Reproducibility

- `git clone https://github.com/pravesh34/AI_generated_text_classification && cd AI_generated_text_classification`
- `python3 -m venv venv && source venv/bin/activate && pip install -r requirements.txt`
- `venv/bin/python train_baseline.py` → writes `models/*.pkl` + `results/baseline_ml_metrics.csv`
- `venv/bin/streamlit run app/app.py` ; `venv/bin/python make_report.py` → this PDF.
- N notebooks: `notebooks/01_*.ipynb` → `06_*.ipynb` execute in order.

References

- Hastie, T., Tibshirani, R., Friedman, J. *Elements of Statistical Learning*. (TF-IDF, LR, SVM).
- Robertson, S. & Zaragoza, H. The Probabilistic Relevance Framework: BM25 & TF-IDF. (2009).
- Mitchell, M. et al. The State and Fate of Linguistic Uncertainty (stylometrics). (2020).
- Reimers, N. & Gurevych, A. Sentence-BERT. (2019). Sanh, T. DistilBERT (2019); Liu, Y. RoBERTa (2019).
- Streamlit, scikit-learn, HuggingFace Transformers documentation (2024–2026).

Appendix Example Predictions (test.csv, random_state 42)

Human (true 0) → LR P(AI) 0.067/0.016/0.003, SVM 0.006/0.001/0.000 — all ✓.
AI (true 1) → LR P(AI) 0.984/0.996/0.9996, SVM 0.9998/1.000/1.000 — all ✓. Ten random rows saved to `results/example_test_set.csv` (5/5 balanced).

File inventory: `models/word_tfidf_vectorizer.pkl` (969K), `char_tfidf_vectorizer.pkl` (1.6M), `logistic_regression_baseline.pkl` (587K), `linear_svm_baseline.pkl` (1.7M), `results/baseline_ml_metrics.csv`, `results/figures/*.png`, `train_baseline.py`, `make_report.py` (this generator).

Report generated automatically by `make_report.py` from live model artifacts and 5,000-row held-out evaluation. All numbers above are computed, not hand-typed.